Project data

Project name	26016 Concourse Viewing Platform
Project number	26016
Author	CaSE
Description	L2-R12-E1_000
Date	11/13/2026
Code	EN

Material

Steel	S 275

Project item SLN73 radial (ends at the ring) - ring SLN55 + SLN56 (continuous)

Name	SLN73 radial (ends at the ring) - ring SLN55 + SLN56 (continuous)
Description	
Analysis	Stress, strain/ simplified loading

Design

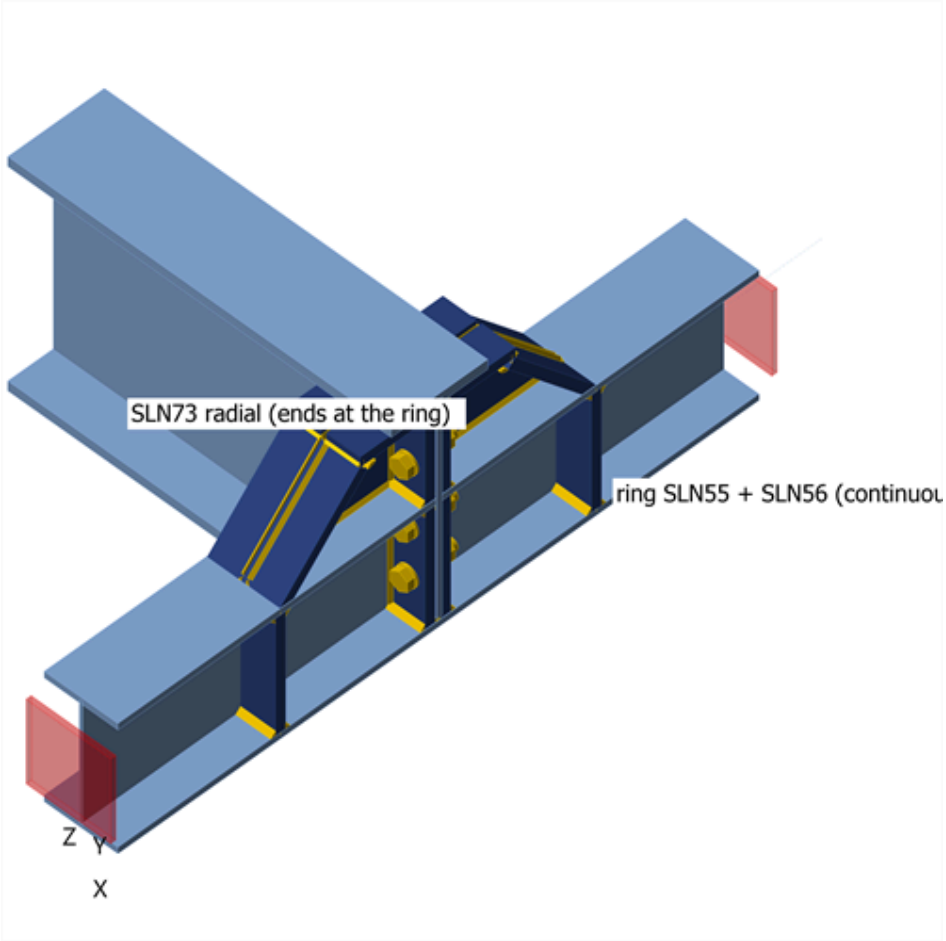
Members

Geometry

Name	Cross-section	β – Direction [°]	γ - Pitch [°]	α - Rotation [°]	Offset ex [mm]	Offset ey [mm]	Offset ez [mm]
SLN73 radial (ends at the ring)	1 - UB457x191x98	0.0	0.0	0.0	76	0	0
ring SLN55 + SLN56 (continuous)	2 - UB254x146x43	-90.0	0.0	0.0	0	0	-104

Supports and forces

Name	Support	Forces in	X [mm]
SLN73 radial (ends at the ring) / begin		Position	0
ring SLN55 + SLN56 (continuous) / begin	N-Vy-Vz-Mx-My-Mz	Position	0
ring SLN55 + SLN56 (continuous) / end	N-Vy-Vz-Mx-My-Mz	Position	0



SLN73 radial (ends at the ring) - ring SLN55 + SLN56 (continuous), Axonometric view

Cross-sections

Name	Material
1 - UB457x191x98	S 275
2 - UB254x146x43	S 275

Bolts

Name	Diameter [mm]	f_y [MPa]	f_u [MPa]	Gross area [mm^2]
M24 10.9	24	900.0	1000.0	452

Load effects (Equilibrium not required)

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
C LC_101	SLN73 radial (ends at the ring) / Begin	2.1	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_102	SLN73 radial (ends at the ring) / Begin	9.2	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_103	SLN73 radial (ends at the ring) / Begin	9.6	-0.1	-86.2	-0.1	-45.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_104	SLN73 radial (ends at the ring) / Begin	7.9	0.0	-86.4	-0.1	-45.4	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_105	SLN73 radial (ends at the ring) / Begin	8.8	-0.1	-86.3	-0.1	-45.4	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_106	SLN73 radial (ends at the ring) / Begin	8.0	0.0	-86.3	-0.1	-45.4	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_107	SLN73 radial (ends at the ring) / Begin	1.8	-0.1	-7.8	0.0	-4.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_108	SLN73 radial (ends at the ring) / Begin	1.7	0.1	-8.4	0.0	-4.6	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_109	SLN73 radial (ends at the ring) / Begin	1.4	0.1	-8.1	0.0	-4.5	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_110	SLN73 radial (ends at the ring) / Begin	0.6	-0.1	-8.1	0.0	-4.4	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_111	SLN73 radial (ends at the ring) / Begin	7.4	-0.1	-61.7	0.0	-32.6	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_112	SLN73 radial (ends at the ring) / Begin	6.3	0.1	-62.3	0.0	-32.8	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_113	SLN73 radial (ends at the ring) / Begin	6.7	0.0	-62.0	0.0	-32.7	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_114	SLN73 radial (ends at the ring) / Begin	5.6	-0.1	-62.0	0.0	-32.7	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_115	SLN73 radial (ends at the ring) / Begin	2.4	-0.1	-13.1	0.0	-7.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_116	SLN73 radial (ends at the ring) / Begin	2.1	0.1	-13.6	0.0	-7.4	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_117	SLN73 radial (ends at the ring) / Begin	2.0	0.0	-13.3	0.0	-7.3	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_118	SLN73 radial (ends at the ring) / Begin	1.1	-0.1	-13.3	0.0	-7.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_119	SLN73 radial (ends at the ring) / Begin	11.2	-0.1	-89.8	-0.1	-47.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_120	SLN73 radial (ends at the ring) / Begin	8.1	0.0	-89.8	-0.1	-47.2	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_121	SLN73 radial (ends at the ring) / Begin	8.5	-0.1	-89.8	-0.1	-47.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_122	SLN73 radial (ends at the ring) / Begin	7.6	0.0	-89.8	-0.1	-47.2	0.2
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_123	SLN73 radial (ends at the ring) / Begin	10.8	-0.1	-68.9	0.0	-36.3	0.1

Name	Member	N [kN]	Vy [kN]	Vz [kN]	Mx [kNm]	My [kNm]	Mz [kNm]
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_124	SLN73 radial (ends at the ring) / Begin	6.7	0.1	-68.9	0.0	-36.3	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_125	SLN73 radial (ends at the ring) / Begin	6.2	0.0	-68.9	0.0	-36.3	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_126	SLN73 radial (ends at the ring) / Begin	4.8	-0.1	-68.9	0.0	-36.3	0.2
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_127	SLN73 radial (ends at the ring) / Begin	5.7	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_128	SLN73 radial (ends at the ring) / Begin	2.5	0.1	-20.2	0.0	-10.9	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_129	SLN73 radial (ends at the ring) / Begin	1.5	0.0	-20.3	0.0	-10.9	-0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_130	SLN73 radial (ends at the ring) / Begin	0.3	-0.1	-20.3	0.0	-10.9	0.2
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_135	SLN73 radial (ends at the ring) / Begin	7.7	0.0	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_141	SLN73 radial (ends at the ring) / Begin	2.3	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_142	SLN73 radial (ends at the ring) / Begin	1.8	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_143	SLN73 radial (ends at the ring) / Begin	2.1	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_144	SLN73 radial (ends at the ring) / Begin	2.0	0.0	-20.3	0.0	-10.9	0.0
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_145	SLN73 radial (ends at the ring) / Begin	10.1	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_146	SLN73 radial (ends at the ring) / Begin	8.3	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_147	SLN73 radial (ends at the ring) / Begin	9.4	-0.1	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_148	SLN73 radial (ends at the ring) / Begin	9.0	0.0	-89.7	-0.1	-47.1	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_151	SLN73 radial (ends at the ring) / Begin	6.9	-0.1	-46.4	0.0	-22.3	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_152	SLN73 radial (ends at the ring) / Begin	9.3	0.0	-63.5	0.0	-35.9	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_153	SLN73 radial (ends at the ring) / Begin	9.2	-0.1	-89.4	0.0	-47.2	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0
C LC_154	SLN73 radial (ends at the ring) / Begin	7.0	-0.1	-46.8	0.0	-22.5	0.1
	SLN73 radial (ends at the ring) / End	0.0	0.0	0.0	0.0	0.0	0.0

Check

Summary

Name	Value	Check status
Analysis	100.0%	OK
Plates	3.0 < 5.0%	OK
Bolts	69.8 < 100%	OK
Welds	100.6 > 100%	**Not OK!**
Buckling	Not calculated	

Plates

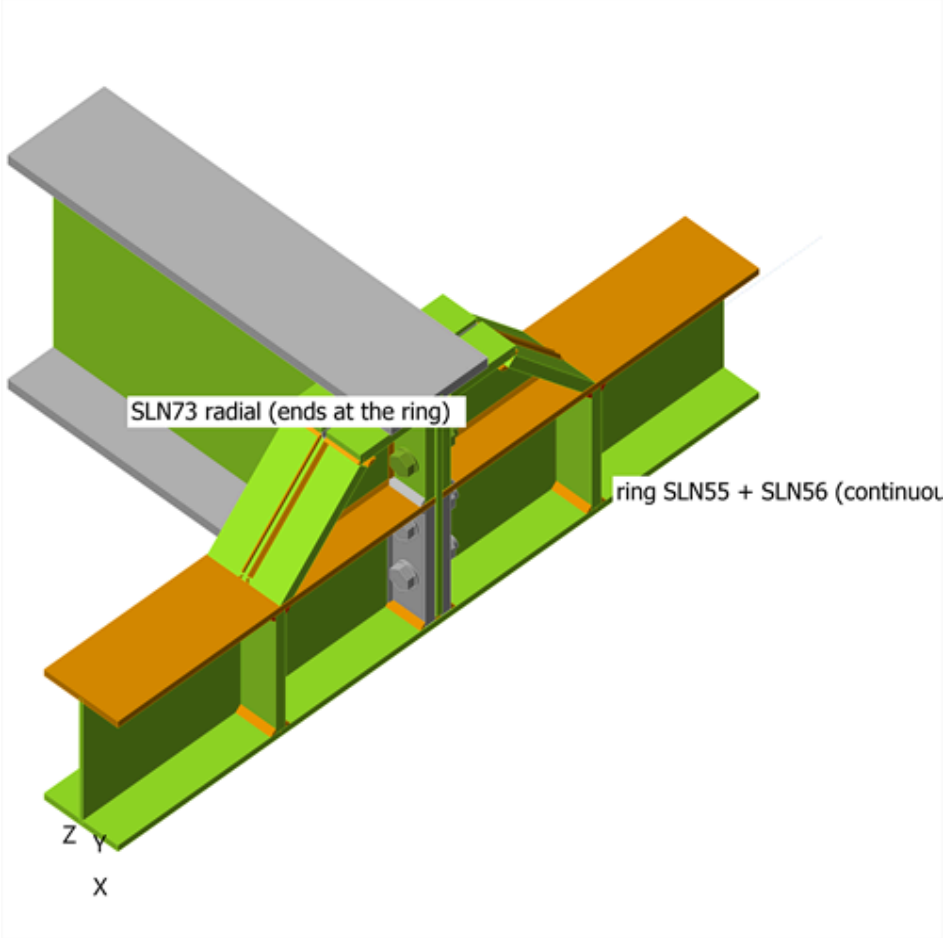
Name	t_p [mm]	Loads	σ_{Ed} [MPa]	ε_{Pl} [%]	$\sigma_{c,Ed}$ [MPa]	Status
SLN73 radial (ends at the ring)-bfl 1	19.6	C LC_122	14.8	0.0	0.0	OK
SLN73 radial (ends at the ring)-tfl 1	19.6	C LC_119	136.9	0.0	5.9	OK
SLN73 radial (ends at the ring)-w 1	11.4	C LC_122	242.8	0.0	70.4	OK
ring SLN55 + SLN56 (continuous)-bfl 1	12.7	C LC_122	280.9	2.8	0.0	OK
ring SLN55 + SLN56 (continuous)-tfl 1	12.7	C LC_122	281.3	3.0	0.0	OK
ring SLN55 + SLN56 (continuous)-w 1	7.2	C LC_122	275.4	0.2	0.0	OK
395x145x20 fin plate upper (SLN55 side)	20.0	C LC_122	275.1	0.1	473.8	OK
395x145x20 fin plate lower outer (SLN55 side)	20.0	C LC_120	211.1	0.0	203.5	OK
395x145x20 fin plate lower inner (SLN55 side)	20.0	C LC_122	211.3	0.0	40.0	OK
395x145x20 fin plate upper (SLN56 side)	20.0	C LC_135	275.1	0.1	470.9	OK
395x145x20 fin plate lower outer (SLN56 side)	20.0	C LC_122	213.1	0.0	204.8	OK
395x145x20 fin plate lower inner (SLN56 side)	20.0	C LC_122	211.2	0.0	39.6	OK
20 ring stiffener at 315 (SLN55 side)	20.0	C LC_122	275.9	0.4	0.0	OK
10 bracket web plate (SLN55 side)	10.0	C LC_122	275.5	0.3	0.0	OK
20 bracket top plate a (SLN55 side)	20.0	C LC_119	275.0	0.0	105.1	OK
20 inclined stiffener a (SLN55 side)	20.0	C LC_122	275.4	0.2	0.0	OK
20 bracket top plate b (SLN55 side)	20.0	C LC_122	275.9	0.4	0.0	OK
20 inclined stiffener b (SLN55 side)	20.0	C LC_122	275.5	0.2	0.0	OK
20 ring stiffener at 315 (SLN56 side)	20.0	C LC_122	275.9	0.4	0.0	OK
10 bracket web plate (SLN56 side)	10.0	C LC_122	275.5	0.3	0.0	OK
20 bracket top plate a (SLN56 side)	20.0	C LC_120	275.9	0.4	0.0	OK
20 inclined stiffener a (SLN56 side)	20.0	C LC_122	275.5	0.2	0.0	OK
20 bracket top plate b (SLN56 side)	20.0	C LC_119	275.0	0.0	105.1	OK
20 inclined stiffener b (SLN56 side)	20.0	C LC_122	275.4	0.2	0.0	OK

Design data

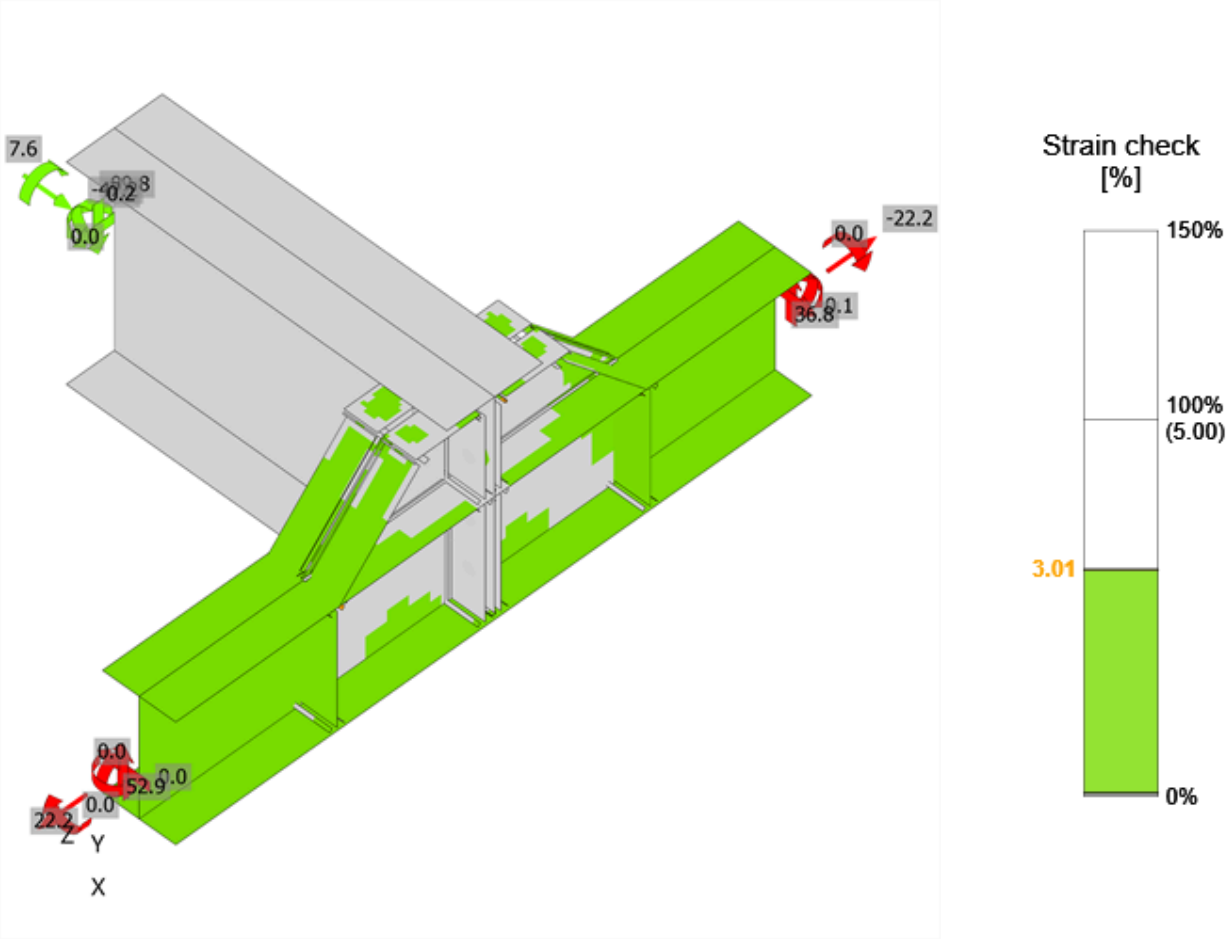
Material	f_y [MPa]	ε_{lim} [%]
S 275	275.0	5.0

Symbol explanation

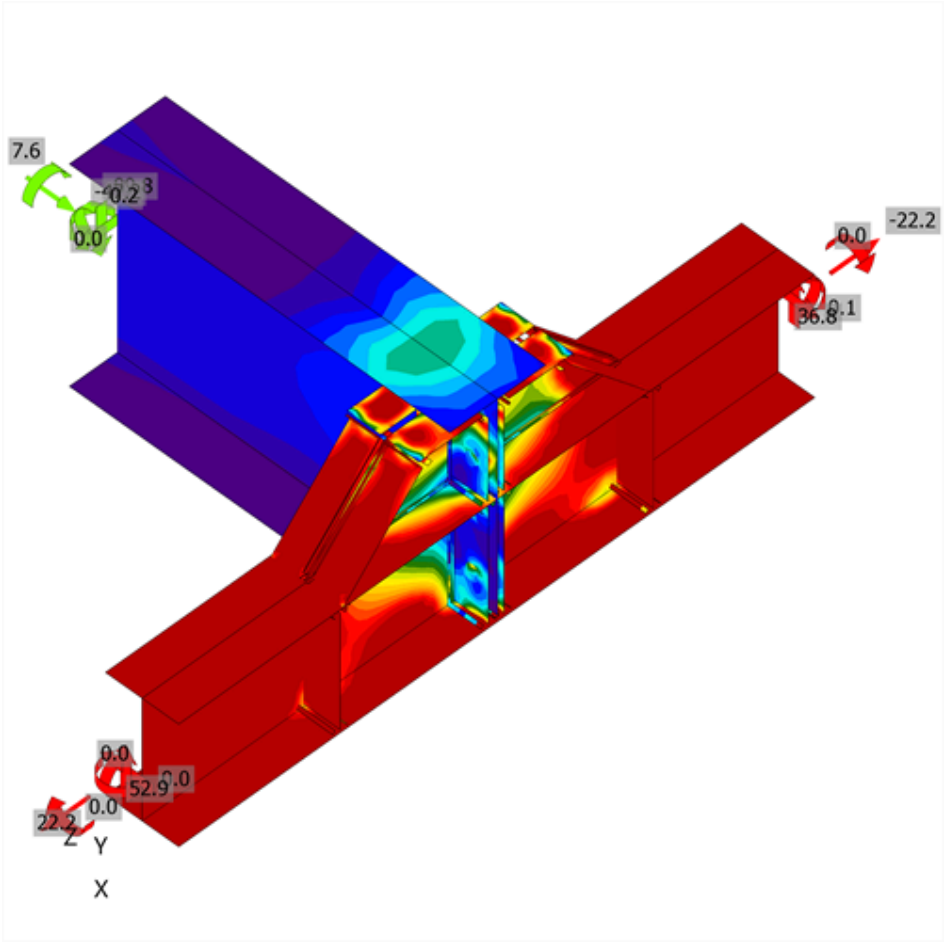
t_p	Plate thickness
σ_{Ed}	Equivalent stress
ε_{Pl}	Plastic strain
$\sigma_{c,Ed}$	Contact stress
f_y	Yield strength
ε_{lim}	Limit of plastic strain



Overall check

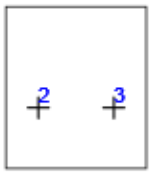
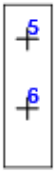
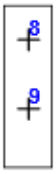


Strain check, C LC_122



Equivalent stress, C LC_122

Bolts

Shape	Item	Grade	Loads	$F_{t,Ed}$ [kN]	$F_{v,Ed}$ [kN]	$F_{b,Rd}$ [kN]	Ut_t [%]	Ut_s [%]	Ut_{ts} [%]	Detailing	Status
	B2	M24 10.9 - 1	C LC_122	46.0	34.5	235.3	18.1	29.2	37.4	OK	OK
	B3	M24 10.9 - 1	C LC_122	6.6	35.2	100.7	2.6	69.8	26.8	OK	OK
	B5	M24 10.9 - 1	C LC_119	3.6	10.2	235.3	1.4	8.7	8.2	OK	OK
	B6	M24 10.9 - 1	C LC_119	1.4	32.3	235.3	0.5	27.4	23.3	OK	OK
	B8	M24 10.9 - 1	C LC_153	0.0	12.7	224.9	0.0	11.2	9.0	OK	OK
	B9	M24 10.9 - 1	C LC_119	12.1	30.9	235.3	4.7	26.2	25.3	OK	OK

Design data

Grade	$F_{t,Rd}$ [kN]	$B_{p,Rd}$ [kN]	$F_{v,Rd}$ [kN]
M24 10.9 - 1	254.2	492.7	141.2

Symbol explanation

$F_{t,Ed}$	Tension force
$F_{v,Ed}$	Resultant of bolt shear forces Vy and Vz in shear planes
$F_{b,Rd}$	Plate bearing resistance - EN 1993-1-8:2005 – Table 3.4
Ut_t	Utilization in tension
Ut_s	Utilization in shear
Ut_{ts}	Interaction of tension and shear - EN 1993-1-8:2005 – Table 3.4
$F_{t,Rd}$	Bolt tension resistance - EN 1993-1-8:2005 – Table 3.4
$B_{p,Rd}$	Punching shear resistance - EN 1993-1-8:2005 – Table 3.4
$F_{v,Rd}$	Bolt shear resistance - EN 1993-1-8:2005 – Table 3.4

Welds

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
10 bracket web plate (SLN55 side)	20 bracket top plate a (SLN55 side)	-	129	-	-	-	-	-	-	-	-	OK	OK
		-	129	-	-	-	-	-	-	-	-	OK	OK
10 bracket web plate (SLN55 side)	20 bracket top plate b (SLN55 side)	-	129	-	-	-	-	-	-	-	-	OK	OK
		-	129	-	-	-	-	-	-	-	-	OK	OK
10 bracket web plate (SLN56 side)	20 bracket top plate a (SLN56 side)	-	129	-	-	-	-	-	-	-	-	OK	OK
		-	129	-	-	-	-	-	-	-	-	OK	OK
10 bracket web plate (SLN56 side)	20 bracket top plate b (SLN56 side)	-	129	-	-	-	-	-	-	-	-	OK	OK
		-	129	-	-	-	-	-	-	-	-	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate upper (SLN55 side)	◢ 7.0 ◣	144	C LC_119	163.6	0.0	75.2	76.5	34.3	40.4	28.3	OK	OK
		◢ 7.0 ◣	144	C LC_119	139.7	0.0	-76.6	59.9	-31.0	34.5	23.1	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	395x145x20 fin plate lower outer (SLN55 side)	◢ 7.0 ◣	69	C LC_122	268.5	0.0	9.4	-120.3	97.6	66.4	47.2	OK	OK
		◢ 7.0 ◣	69	C LC_120	397.0	0.2	-56.7	-4.4	-226.8	98.1	64.3	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower outer (SLN55 side)	◢ 7.0 ◣	68	C LC_119	113.2	0.0	45.3	59.3	-8.5	28.0	21.2	OK	OK
		◢ 7.0 ◣	69	C LC_119	145.9	0.0	85.1	-68.3	-4.0	36.1	26.8	OK	OK
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower outer (SLN55 side)	◢ 4.2 ◣	233	C LC_119	116.1	0.0	1.5	-1.4	67.0	28.7	20.7	OK	OK
		◢ 4.2 ◣	232	C LC_122	118.8	0.0	-31.7	9.1	65.4	29.3	20.4	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	395x145x20 fin plate lower inner (SLN55 side)	◢ 7.0 ◣	69	C LC_120	396.6	0.0	36.7	80.9	213.2	98.0	51.4	OK	OK
		◢ 7.0 ◣	69	C LC_122	397.1	0.3	50.2	-32.8	-225.1	98.1	55.6	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower inner (SLN55 side)	◢ 7.0 ◣	69	C LC_119	120.1	0.0	-17.5	-68.6	1.6	29.7	22.2	OK	OK
		◢ 7.0 ◣	69	C LC_119	146.3	0.0	-108.0	56.9	1.2	36.1	25.2	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower inner (SLN55 side)	◢ 4.2 ◣	232	C LC_119	95.3	0.0	-5.6	-2.7	54.8	23.5	16.8	OK	OK
		◢ 4.2 ◣	232	C LC_122	154.2	0.0	-17.9	18.4	86.5	38.1	24.4	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate upper (SLN56 side)	◢ 7.0 ◣	144	C LC_119	140.9	0.0	-76.4	-60.4	31.9	34.8	23.2	OK	OK
		◢ 7.0 ◣	144	C LC_119	166.0	0.0	76.9	-77.4	-35.1	41.0	28.7	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	395x145x20 fin plate lower outer (SLN56 side)	◢ 7.0 ◣	69	C LC_122	397.0	0.2	-56.6	4.4	226.8	98.1	64.4	OK	OK
		◢ 7.0 ◣	69	C LC_122	268.4	0.0	9.4	120.8	-96.9	66.3	47.2	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower outer (SLN56 side)	◢ 7.0 ◣	69	C LC_119	146.1	0.0	84.7	68.6	3.3	36.1	26.9	OK	OK
		◢ 7.0 ◣	68	C LC_119	115.9	0.0	47.2	-60.5	8.7	28.6	21.7	OK	OK
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower outer (SLN56 side)	◢ 4.2 ◣	232	C LC_122	118.5	0.0	-32.2	-9.7	-65.1	29.3	20.3	OK	OK
		◢ 4.2 ◣	233	C LC_119	116.1	0.0	1.6	1.3	-67.0	28.7	20.6	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	395x145x20 fin plate lower inner (SLN56 side)	◢ 7.0 ◣	69	C LC_122	397.1	0.3	50.2	32.7	225.1	98.1	55.6	OK	OK
		◢ 7.0 ◣	69	C LC_122	396.6	0.0	37.1	-81.4	-213.0	98.0	51.5	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	395x145x20 fin plate lower inner (SLN56 side)	◢ 7.0 ◣	69	C LC_119	146.7	0.0	-107.7	-57.4	-1.6	36.2	25.3	OK	OK
		◢ 7.0 ◣	69	C LC_119	122.3	0.0	-19.4	69.7	-1.3	30.2	22.6	OK	OK
ring SLN55 + SLN56 (continuous)-w 1	395x145x20 fin plate lower inner (SLN56 side)	◢ 4.2 ◣	232	C LC_122	153.5	0.0	-18.3	-18.8	-86.0	37.9	24.5	OK	OK
		◢ 4.2 ◣	232	C LC_122	95.6	0.0	18.5	-19.1	50.6	23.6	16.6	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	20 ring stiffener at 315 (SLN55 side)	◢ 7.0 ◣	146	C LC_119	396.9	0.2	91.6	129.1	-181.8	98.1	85.9	OK	OK
		◢ 7.0 ◣	146	C LC_120	338.4	0.0	-108.3	41.1	-180.5	83.6	57.7	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
ring SLN55 + SLN56 (continuous)-tfl 1	20 ring stiffener at 315 (SLN55 side)	◢ 7.0 ◣	146	C LC_153	396.8	0.1	153.0	-209.5	-27.8	98.0	85.4	OK	OK
		◢ 7.0 ◣	146	C LC_153	407.2	6.5	-203.4	199.6	40.3	100.6	99.3	OK	**Not OK!**
ring SLN55 + SLN56 (continuous)-tfl 1	10 bracket web plate (SLN55 side)	◢ 4.2 ◣	288	C LC_122	397.3	0.4	-47.9	-37.1	-224.6	98.2	60.8	OK	OK
		◢ 4.2 ◣	288	C LC_122	397.4	0.5	18.4	-8.6	-229.0	98.2	68.8	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener a (SLN55 side)	◢ 7.0 ◣	67	C LC_119	402.8	3.8	171.9	-184.4	101.2	99.5	96.9	OK	OK
		◢ 7.0 ◣	67	C LC_122	399.9	2.0	148.6	193.9	-91.3	98.8	86.0	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener b (SLN55 side)	◢ 7.0 ◣	67	C LC_119	402.6	3.7	-142.6	189.5	106.7	99.5	92.8	OK	OK
		◢ 7.0 ◣	67	C LC_122	401.2	2.8	-141.1	-200.2	-83.3	99.1	84.2	OK	OK
ring SLN55 + SLN56 (continuous)-bfl 1	20 ring stiffener at 315 (SLN56 side)	◢ 7.0 ◣	146	C LC_119	396.9	0.2	91.1	128.4	182.4	98.1	85.5	OK	OK
		◢ 7.0 ◣	146	C LC_122	339.2	0.0	-108.7	41.3	180.9	83.8	59.9	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 ring stiffener at 315 (SLN56 side)	◢ 7.0 ◣	146	C LC_153	396.8	0.1	151.7	-209.8	28.0	98.0	85.5	OK	OK
		◢ 7.0 ◣	146	C LC_119	407.2	6.5	-203.8	199.6	-39.7	100.6	99.3	OK	**Not OK!**
ring SLN55 + SLN56 (continuous)-tfl 1	10 bracket web plate (SLN56 side)	◢ 4.2 ◣	288	C LC_122	397.4	0.5	18.4	8.4	229.0	98.2	68.7	OK	OK
		◢ 4.2 ◣	288	C LC_122	397.3	0.4	-47.8	36.8	224.7	98.2	60.8	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener a (SLN56 side)	◢ 7.0 ◣	67	C LC_119	402.7	3.7	-143.4	189.5	-106.2	99.5	94.4	OK	OK
		◢ 7.0 ◣	67	C LC_122	401.2	2.8	-141.5	-200.0	83.4	99.1	80.7	OK	OK
ring SLN55 + SLN56 (continuous)-tfl 1	20 inclined stiffener b (SLN56 side)	◢ 7.0 ◣	67	C LC_119	402.9	3.9	172.9	-184.4	-100.7	99.5	96.5	OK	OK
		◢ 7.0 ◣	67	C LC_122	399.9	2.0	149.2	193.8	91.3	98.8	88.6	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
395x145x20 fin plate upper (SLN55 side)	10 bracket web plate (SLN55 side)	◢ 4.2 ◣	167	C LC_119	399.3	1.7	-302.5	109.5	103.2	98.7	76.5	OK	OK
		◢ 4.2 ◣	167	C LC_119	400.2	2.2	133.2	187.4	111.1	98.9	80.6	OK	OK
395x145x20 fin plate upper (SLN56 side)	10 bracket web plate (SLN56 side)	◢ 4.2 ◣	167	C LC_119	400.2	2.2	133.6	-187.5	-110.8	98.9	80.3	OK	OK
		◢ 4.2 ◣	167	C LC_119	399.3	1.7	-301.7	-110.1	-103.5	98.7	76.6	OK	OK
20 bracket top plate a (SLN55 side)	20 inclined stiffener a (SLN55 side)	◢ 7.0 ◣	67	C LC_122	397.4	0.5	-48.4	3.0	227.7	98.2	78.1	OK	OK
		◢ 7.0 ◣	67	C LC_122	396.6	0.0	-55.7	16.1	226.2	98.0	67.4	OK	OK
20 bracket top plate b (SLN55 side)	20 inclined stiffener b (SLN55 side)	◢ 7.0 ◣	67	C LC_122	397.1	0.3	6.6	-33.4	226.8	98.1	76.9	OK	OK
		◢ 7.0 ◣	67	C LC_122	379.1	0.0	-66.8	30.6	213.3	93.7	66.2	OK	OK
20 bracket top plate a (SLN56 side)	20 inclined stiffener a (SLN56 side)	◢ 7.0 ◣	67	C LC_122	397.1	0.3	6.3	-33.4	-226.8	98.1	73.3	OK	OK
		◢ 7.0 ◣	67	C LC_122	379.0	0.0	-65.9	29.8	-213.4	93.6	60.9	OK	OK
20 bracket top plate b (SLN56 side)	20 inclined stiffener b (SLN56 side)	◢ 7.0 ◣	67	C LC_122	397.4	0.5	-48.5	2.9	-227.7	98.2	76.6	OK	OK
		◢ 7.0 ◣	67	C LC_122	396.6	0.0	-56.8	16.7	-226.0	98.0	69.7	OK	OK
10 bracket web plate (SLN55 side)	20 inclined stiffener a (SLN55 side)	◢ 4.2 ◣	222	C LC_119	398.4	1.1	-262.2	-147.0	-91.5	98.4	92.3	OK	OK
		◢ 4.2 ◣	222	C LC_119	398.5	1.2	-80.7	189.2	122.3	98.5	78.9	OK	OK
10 bracket web plate (SLN55 side)	20 inclined stiffener b (SLN55 side)	◢ 4.2 ◣	222	C LC_119	398.8	1.3	128.1	121.5	-181.0	98.5	93.5	OK	OK
		◢ 4.2 ◣	222	C LC_119	398.1	0.9	165.7	-174.7	114.8	98.4	69.6	OK	OK
10 bracket web plate (SLN56 side)	20 inclined stiffener a (SLN56 side)	◢ 4.2 ◣	222	C LC_119	398.8	1.4	128.5	121.7	180.8	98.5	93.3	OK	OK
		◢ 4.2 ◣	222	C LC_119	398.1	0.9	165.5	-174.6	-114.9	98.4	75.0	OK	OK

Item	Edge	T_w [mm]	L [mm]	Loads	$\sigma_{w,Ed}$ [MPa]	ε_{Pl} [%]	$\sigma_{\perp}$ [MPa]	$\tau_{\perp}$ [MPa]	$\tau_{\parallel}$ [MPa]	Ut [%]	Ut_c [%]	Detailing	Status
10 bracket web plate (SLN56 side)	20 inclined stiffener b (SLN56 side)	◢ 4.2 ◣	222	C LC_119	398.4	1.1	-261.9	-147.1	91.6	98.4	90.9	OK	OK
		◢ 4.2 ◣	222	C LC_119	398.5	1.2	-80.9	189.1	-122.4	98.5	77.4	OK	OK
395x145x20 fin plate upper (SLN55 side)	20 bracket top plate a (SLN55 side)	◢ 4.2 ◣	67	C LC_122	396.9	0.2	-206.2	-185.7	-62.0	98.1	70.6	OK	OK
		◢ 4.2 ◣	67	C LC_122	396.9	0.2	228.2	-165.7	-87.8	98.1	78.5	OK	OK
395x145x20 fin plate upper (SLN55 side)	20 bracket top plate b (SLN55 side)	◢ 4.2 ◣	67	C LC_122	402.5	3.6	-231.7	190.0	3.5	99.5	96.1	OK	OK
		◢ 4.2 ◣	67	C LC_122	402.5	3.6	-221.5	189.7	-40.9	99.5	94.3	OK	OK
395x145x20 fin plate upper (SLN56 side)	20 bracket top plate a (SLN56 side)	◢ 4.2 ◣	67	C LC_120	402.2	3.4	-232.1	189.6	-2.4	99.4	95.9	OK	OK
		◢ 4.2 ◣	67	C LC_120	402.2	3.5	-222.1	189.0	41.7	99.4	92.3	OK	OK
395x145x20 fin plate upper (SLN56 side)	20 bracket top plate b (SLN56 side)	◢ 4.2 ◣	67	C LC_122	396.9	0.2	-207.7	-184.9	62.6	98.1	71.1	OK	OK
		◢ 4.2 ◣	67	C LC_122	396.9	0.2	228.7	-165.3	88.1	98.1	76.3	OK	OK

Design data

Material	f_u [MPa]	β_w [-]	$\sigma_{w,Rd}$ [MPa]	0.9 σ [MPa]
S 275	430.0	0.85	404.7	309.6

Symbol explanation

T_w	Throat thickness a
L	Length
$\sigma_{w,Ed}$	Equivalent stress
ε_{Pl}	Strain
$\sigma_{\perp}$	Perpendicular stress
$\tau_{\perp}$	Shear stress perpendicular to weld axis
$\tau_{\parallel}$	Shear stress parallel to weld axis
Ut	Utilization
Ut_c	Weld capacity estimation
◢	Fillet weld
f_u	Ultimate strength of weld
β_w	Correlation factor - EN 1993-1-8:2005 – Table 4.1
$\sigma_{w,Rd}$	Equivalent stress resistance
0.9 σ	Perpendicular stress resistance: 0.9*fu/γM2

Buckling

Buckling analysis was not calculated.

Code settings

Item	Value	Unit	Reference
Safety factor γ_{M0}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M1}	1.00	-	EN 1993-1-1:2005 – 6.1
Safety factor γ_{M2}	1.25	-	EN 1993-1-1:2005 – 6.1, EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{M3}	1.25	-	EN 1993-1-8:2005 – Table 2.1
Safety factor γ_{C}	1.50	-	EN 1992-1-1:2004 – 2.4.2.4
Safety factor γ_{Inst}	1.20	-	EN 1992-4:2018 – Table 4.1
Joint coefficient βj	0.67	-	EN 1993-1-8:2005 – 6.2.5(7)
Effective area - influence of mesh size	0.10	-	
Friction coefficient - concrete	0.25	-	EN 1993-1-8:2005
Friction coefficient in slip-resistance	0.30	-	EN 1993-1-8:2005 – Table 3.7
Limit plastic strain	0.05	-	EN 1993-1-5:2006
Detailing	Yes		
Distance between bolts [d0]	2.20	-	EN 1993-1-8:2005 – Table 3.3
Distance between bolts and edge [d0]	1.20	-	EN 1993-1-8:2005 – Table 3.3
Concrete breakout resistance check	Both		
Cracked concrete	Yes		EN 1992-4:2018
Local deformation check	Yes		
Local deformation limit	0.03	-	CIDECT DG 1, 3 – 1.1
Geometrical nonlinearity (GMNA)	Yes		Analysis with large deformations for hollow section joints
Braced system	No		EN 1993-1-8:2005 – 5.2.2.5